medcoder — Auditable Medical-Coding Pipeline

AppliedAI · Opus AI Engineer · Exercise 2

Design Document (1–2 pp distillation)

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1 · Architecture & data flow

Design thesis. The LLM is a *constrained reasoning component inside a deterministic, observable pipeline* — not the pipeline itself. We make a stochastic model auditable by **pre-constraining it with retrieval** (it can only choose real codes), **independently verifying it with an auditor agent**, and **post-constraining it with a deterministic rule engine** grounded in the ICD-10-CM Official Guidelines.

The pipeline has seven stages, each idempotent and independently retryable:

```
 note.txt ─► ① ingest ─► ② extract ─► ③ retrieve ─► ④ code ─► ⑤ audit ─► ⑥ rules ─► ⑦ assemble ─►
CodingResult.json
              (det.)      (LLM-A)       (det.)      (LLM-A)    (LLM-B)      (det.)        (det.)
```

1. **Ingest** — normalize, segment SOAP sections, window long multi-page notes with overlap, preserve **global character offsets**. Encounter type (inpatient vs outpatient — it governs whether “probable / suspected” diagnoses are codable per ICD-10-CM Guideline IV.H) is classified by the extraction LLM (§3), with a deterministic keyword heuristic as fallback.
2. **Extract (LLM-A).** Returns `ExtractedFact[]` — verbatim span, normalized clinical term, **assertion status** (present / absent / possible / hypothetical / family / historical), kind (dx / px / symptom). A deterministic NegEx/ConText-style backstop overrules clear LLM polarity slips (“denies chest pain” must never be coded as active chest pain).
3. **Retrieve** — hybrid search per fact (see §2).
4. **Code (LLM-A).** Picks 1–N codes **only from the retrieved whitelist**, with verbalised confidence and a rationale that quotes the evidence span.
5. **Audit (LLM-B independent).** Re-reads each (evidence, code) pair against the cited evidence + code description and returns agree/disagree + note. *Selective + batched*: only procedures + low-confidence diagnoses go through the heavy auditor.
6. **Rules (deterministic).** Excludes1 conflicts, unspecified-when-specific, missing 7th character on injury/pregnancy chapters, evidence anchoring, procedure-without-supporting-diagnosis, format validity. Emits typed `Warning[]` (missing_information / ambiguity / conflict).
7. **Assemble (deterministic).** Blends and tiers the confidence (§3), builds the Pydantic-validated `CodingResult` and a `RunMetadata` envelope with `trace_id`, model snapshots, prompt version, config hash, and per-stage metrics.

2 · Code retrieval / filtering strategy

The single biggest failure mode for full-vocabulary ICD-10 coding is the LLM **hallucinating non-existent codes** — across GPT-4/Gemini/Llama models, up to 35% of generated ICD-10 codes are non-billable or invalid (NEJM AI, 2024). We eliminate this structurally: the LLM never free-generates a code.

For each extracted fact, retrieval queries its `normalized_term` **plus a few LLM-emitted synonyms** (`query_terms`, e.g. “MI” → “myocardial infarction”) and **merges** the results — a lightweight *query expansion* that widens recall against vocabulary mismatch without touching the whitelist. Each query runs:

- **Dense search** with `sentence-transformers/all-MiniLM-L6-v2` over a FAISS inner-product index (catches paraphrases — “DM type 2” finds “type 2 diabetes mellitus”). Returns top-N=50.
- **Lexical BM25** over the same descriptions (catches exact clinical phrasing — “essential hypertension” matters word-for-word). Returns top-N=50.
- **Reciprocal Rank Fusion** (Cormack et al. 2009): each retriever contributes `1 / (k + rank)`; fused scores are summed. RRF needs **no score calibration** between heterogeneous scorers — the empirically dominant hybrid fusion choice. Candidates from all query terms are merged (best score per code); the **top-K=15 become the whitelist**, each carrying its post-merge *fused rank* (which feeds the confidence blend, §3).

That whitelist is a **hard constraint**, not soft context. The coder agent’s response is re-validated against the candidate set; any out-of-list code is dropped and logged. This makes hallucinated codes *structurally impossible*.

Data: we bundle the **real CDC ICD-10-CM FY2027 catalog** (~74,879 codes, US public domain) — using a synthetic ICD-10 catalog would dodge the large-code-space challenge the exercise tests. CPT is **AMA-copyrighted with no free tier**, so we ship a clearly-marked **synthetic CPT-shaped catalog** and make licensed real CPT a one-config-line drop-in.

Production swap (documented): the same `Retriever` interface backs equally well by **Postgres `pgvector` + `tsvector` + `pg_trgm`** — one datastore for semantic + full-text + fuzzy in one operational footprint.

3 · LLM usage & prompting approach

Three role-specialized agents (mirroring the real-world coder → QA-auditor workflow), all behind one **LiteLLM** Chat Completions gateway so providers (and per-agent models) swap by env var:

Agent	Human analog	Model (default)	Constraint
Extraction	clinical scribe	`openai/gpt-5.4-mini`	reason-then-format; assertion regex
Coder	medical coder	`openai/gpt-5.4-mini`	may only choose from whitelist
Auditor	QA auditor	`anthropic/claude-haiku-4-5-20251001`	**different model family** by default

The **auditor defaulting to a different model family** is deliberate — research shows heterogeneous verifiers cut correlated errors and self-preference bias (arXiv 2410.21819; A-HMAD 2025). Same-model fallback is supported but flagged as weaker. **Selective verification** (only procedures + low-confidence diagnoses) and **batched calls** (one extract / one code / one audit per note) keep the cost discipline reasonable. The extraction call does triple duty in one shot (no added cost): facts, a note-level **`encounter_type`** (replacing brittle keyword counting; §1), and per-fact **`query_terms`** synonyms that feed retrieval query expansion (§2).

Prompting choices:

- *Reason-then-format.* Strict JSON-only constraints degrade reasoning quality (arXiv 2408.02442). Prompts instruct the model to think internally first, then emit a single JSON object validated against a Pydantic schema. On validation failure we **repair-retry** once with the error appended to the conversation — almost always sufficient.

- *Versioned prompt files* (`prompts/extraction_p2.txt` , etc.). Prompt versions are part of the `config_hash` so a prompt change is visible in the audit log.
- *Pinned model IDs + bounded sampling* (e.g. `openai/gpt-5.4-mini`): `temp=0` where honoured (Claude); GPT-5 reasoning models reject it, so determinism rests on Structured Outputs + low `reasoning_effort` (hashed; self-consistency off).

Confidence we surface ≠ raw LLM confidence. Verbalised LLM confidence is systematically overconfident (Xiong et al. 2023). We blend three signals — fused retrieval rank (post-merge rank, not raw RRF score), coder confidence (discounted), and an auditor adjustment (+0.15 for agree, −0.30 for disagree) — and bin into 🟢 / 🟡 / 🔴 tiers using gold-tuned thresholds. (Formal isotonic/Platt calibration is wired as an extension — needs a larger labelled set than the demo gold supports.)

4 · Key decisions & trade-offs

Decision	Why	Trade-off accepted
Retrieve-then-constrain (whitelist, not free generation)	Eliminates hallucinated codes; turns a 75k-way generation problem into a k-way selection (6% → ~100% on a comparable task — arXiv 2407.12849)	Bounded by **retriever recall@k** — the upstream recall ceiling on retrievable codes (not computed in the eval)
Coder + independent Auditor (multi-LLM)	Best precision/recall on MIMIC-IV (MDPI Informatics 2026); heterogeneous verifiers cut correlated errors (A-HMAD 2025)	Extra LLM calls — mitigated by selective+batched verification
Hybrid retrieval + RRF	Exact terms *and* paraphrases; no score-scale tuning	Two indexes to build; ~1 minute on 75k codes
Bounded 3-agent decomposition (no swarm)	MAST (NeurIPS 2025) shows speculative swarms add latency and “silent gray errors” without reliable accuracy gains	Foregoes ensemble-style accuracy gains for predictability
Real ICD-10 + synthetic CPT	Solves the brief’s large-code-space challenge; legal under AMA licensing	CPT demo runs on synthetic codes
LiteLLM gateway	Multi-provider in one call; built-in `mock_response` for keyless tests	Thin dependency
temp=0 + pinned snapshots over sampling	Reproducibility / auditability is an explicit requirement	Foregoes ~1–3 pt accuracy from self-consistency (offered as optional)
Deterministic rule engine for hard checks	Auditable, exact, no LLM cost	Full guideline coverage needs the ICD-10 tabular XML + CMS NCCI
CLI-only core (FastAPI/UI as extensions)	Matches the offline brief; no over-build	No reviewer UI — but the JSON contract supports override in-place
Blended-then-tiered confidence	Raw LLM confidence is systematically overconfident	Calibration is gold-tuned, not formally calibrated (extension)

5 · Limitations & extensions

Limitations (honest). *Assistive, not autonomous* — SOTA full-vocabulary ICD-10 tops out around micro-F1 ≈ 0.54 (RAG-Coding, 2026); a human reviewer is required by design and the payload (evidence spans, tiered confidence, mutable reviewer fields) is shaped to make that review fast. *CPT is synthetic* — real-CPT accuracy must be re-validated on a licensed catalog (architecture is drop-in). *General-purpose embedder* (`all-MiniLM-L6-v2`) is a demo compromise; production choice is biomedical (SapBERT / PubMedBERT). *Rule engine is a curated subset* (format, billable, Excludes1 short list, 7th-character chapters, dx↔px linkage, evidence anchoring); full guideline coverage needs the ICD-10 tabular XML + CMS NCCI refresh pipeline. *Confidence calibration* is gold-tuned thresholds, not formal Platt / isotonic — needs a larger labelled set. *Reproducibility* is engineered (temp=0, pinned dated snapshots, versioned prompts, full audit log) but not bit-for-bit guaranteed across provider model updates — exactly why we pin and log everything. *Evaluation* is directional on a small (n=4) authored gold set — ICD-10 micro-F1 ≈ 0.51 (≈ the ~0.54 SOTA ceiling), CPT micro-F1 ≈ 0.75; the metric methodology (`scripts/evaluate.py` : micro P/R/F1 for ICD and CPT, exact-match ratio, ICD-10-hierarchical micro-F1) is sound — only the sample size is small. *Precision (≈ 0.40) is over-coding-bound* — the coder emits more codes than gold — so a more selective coder and a larger gold set are the precision lever, not retrieval; the v2 changes (query expansion + LLM encounter type) lifted recall at held precision (ICD recall 0.62 → 0.71, encounter-type 3/4 → 4/4).

Extensions (designed for, not built). Postgres hybrid retrieval (`pgvector` + `tsvector` + `pg_trgm`); biomedical embeddings + SNOMED→ICD crosswalk; full tabular-rule + NCCI engine; FastAPI + reviewer UI; self-consistency confidence; provider-query drafting; licensed real CPT (one config line); LLM-observability (Langfuse/OpenTelemetry via LiteLLM callbacks). Stages are idempotent → map to Airflow/Celery for orchestration.

References (load-bearing). NEJM AI 2024 (LLMs are poor coders); arXiv 2407.12849 (retrieve-then-rerank); MDPI Informatics 2026 (coder+auditor); MAST/NeurIPS 2025; Cormack 2009 (RRF); ICD-10-CM Official Guidelines FY2026 — full list in `docs/Plan.md` §20.